The basic primitive recursive functions are lambda definable

The Open Logic Project contributors

openlogic:lambda-calculus/lambda-definability:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author lemma labelled lem:basic (1094-1098); the proof span 1100-1108 is the author proof environment that immediately follows it.

Editorial note. The companion lemma lem:comp is not delivered: its proof defers the defining reduction to an exercise (rejected entry in the lane's rejected.json).

Editorial scope. One original result of the Open Logic Project unit lambda-calculus/lambda-definability, counted once: lambda definability of the basic primitive recursive functions. Only the counted lemma and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- Lambda definability. An n-ary number-theoretic function f is lambda definable when there is a lambda term H such that for all n-tuples, H applied to the Church numerals of the arguments beta-reduces to the Church numeral of the value.. It is the criterion the displayed terms must meet (1100-1108).
- Church numerals. The Church numeral for n is the term applying its first argument n times to its second argument.. The displayed Succ and projection terms act on Church numerals by iteration (1100-1108).

Lemma 0.1. *The basic primitive recursive functions* 0, S, *and projections* proj *ni are !!lambda definable.*

Proof. They are !!lambda defined by the following terms:

$$\begin{aligned} \text{fn}\, Zero\, \text{id}\, \lambda[a][\lambda[fx][x]] \\ \text{fn}\, Succ\, \text{id}\, \lambda[a][\lambda[fx][f(afx)]] \\ \text{fn}\, Proj_i^n\, \text{id}\, \lambda[x_0 \dots x_{n-1}][x_i] \end{aligned}$$

□

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